

MTL106: Probability and Stochastic Processes

Lecture 10

January 23, 2026

Characteristic Function: Let X be a random variable. The complex-valued function ϕ defined on $\mathbb{R}$ by

$$\phi(t) = E[e^{itX}] = E(\cos tX) + iE(\sin tX), \quad t \in \mathbb{R},$$

where $i = \sqrt{-1}$ is the imaginary unit, is called the *characteristic function* (CF) of random variable X . Clearly, for the *discrete case*,

$$\phi(t) = \sum_k (\cos tx_k + i \sin tx_k) P(X = x_k),$$

and for the *continuous case*,

$$\phi(t) = \int_{-\infty}^{\infty} \cos tx \, f(x) dx + i \int_{-\infty}^{\infty} \sin tx \, f(x) dx.$$

Example 1. Let X be a random variable with PDF

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right), \quad x \in \mathbb{R}.$$

Then the CF is given by

$$\begin{aligned} \phi(t) &= \int_{-\infty}^{\infty} e^{itx} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{x^2 - 2itx}{2}\right\} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{(x - it)^2 + t^2}{2}\right\} dx \\ &= \frac{e^{-t^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{(x - it)^2}{2}\right\} dx \\ &= \frac{e^{-t^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{z^2}{2}\right\} dz \quad [x - it = z] \\ &= e^{-t^2/2}, \quad t \in \mathbb{R} \quad \left[\text{since } \int_{-\infty}^{\infty} e^{-z^2/2} dz = \sqrt{2\pi}\right]. \end{aligned}$$

Remark 1. Unlike MGF (that may not exist for some distributions), a CF always exists, which makes it a much more convenient tool. In fact, it is easy to see that ϕ is continuous on $\mathbb{R}$, $|\phi(t)| \leq 1 \quad \forall \, t$, and $\phi(-t) = \overline{\phi(t)}$, where $\overline{\phi}$ is the complex conjugate of ϕ . Thus $\overline{\phi}$ is the CF of $-X$. Moreover, ϕ uniquely determines the CDF of random variable X .

Some Moment Inequalities:

Theorem 1. Let $h(X)$ be a nonnegative Borel-measurable function of a random variable X . If $E[h(X)]$ exists, then for every $\epsilon > 0$,

$$P\{h(X) \geq \epsilon\} \leq \frac{E[h(X)]}{\epsilon}. \quad (1)$$

Proof. The result can be proved for both discrete and continuous random variables.

1. *When X is discrete.* Let $P(X = x_k) = p_k$, $k = 1, 2, \dots$, Then

$$E[h(X)] = \sum_k h(x_k)p_k = \left\{ \sum_A + \sum_{A^c} \right\} h(x_k)p_k,$$

where $A = \{k : h(x_k) \geq \epsilon\}$. Then

$$E[h(X)] \geq \sum_A h(x_k)p_k \geq \epsilon \sum_A p_k = \epsilon P\{h(X) \geq \epsilon\}.$$

2. *When X is continuous.* Proof is very trivial as one can replace the summation by integration in the aforementioned derivation.

Markov's Inequality: In the aforementioned inequality (1), let us put $h(X) = |X|^r$ and $\epsilon = K^r$, where $r > 0$ and $K > 0$. Then

$$P\{|X| \geq K\} \leq \frac{E[|X|^r]}{K^r}, \quad (2)$$

which is known as *Markov's Inequality*.

Example 2. The number of customers arrive at a departmental store per hour with a mean of 6. What is the probability that more than 12 customers visit the store in an hour?

Solution. Let X be a random variable which denotes the number of customers. Given that $E(X) = 6$. Using Markov's inequality, we have

$$P(X \geq 12) \leq \frac{6}{12} = 0.5.$$

Hence the probability of more than 12 customers visit the store in an hour will be maximum 0.5.

Chebychev's Inequality: In inequality (1), if we put $h(X) = (X - \mu)^2$ and $\epsilon = K^2\sigma^2$, we get

$$P\{|X - \mu| \geq K\sigma\} \leq \frac{1}{K^2}, \quad (3)$$

where $E(X) = \mu$, and $Var(X) = \sigma^2$. The inequality (3) is known as the *Chebychev's Inequality*. One can also write (3) as

$$P\{|X - \mu| < K\sigma\} \geq 1 - \frac{1}{K^2}.$$

or

$$P\{|X - \mu| \geq K\} \leq \frac{\sigma^2}{K^2}.$$

or

$$P\{|X - \mu| < K\} \geq 1 - \frac{\sigma^2}{K^2}.$$

Remark 1. If we wish to be consistent with our definition of a CDF as $F(x) = P(X \leq x)$, then we may reformulate (1) in the following form:

$$P\{h(X) > \epsilon\} < \frac{E[h(X)]}{\epsilon}.$$

Remark 2. For random variables with finite second-order moments, one cannot do better than the Chebychev's inequality.

Example 3. Let X be distributed with PDF

$$f(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then, $E(X) = 1/2$, $E(X^2) = 1/3$, and $Var(X) = 1/12$. We have

$$P\{|X - 1/2| < 2\sqrt{1/12}\} = P\{1/2 - 1/\sqrt{3} < X < 1/2 + 1/\sqrt{3}\} = 1.$$

However, from Chebychev's inequality

$$P\{|X - 1/2| < 2\sqrt{1/12}\} \geq 1 - 1/4 = 0.75.$$

Remark 3. It is possible to improve upon Chebychev's inequality, at least in some cases, if we assume the existence of higher-order moments.

Remark 4. Chebyshev's inequality is an empirical rule that allows us to predict proportion of observations that is likely to lie between an interval defined using mean and standard deviation.

Example 4. Amount spent per month by a segment of credit card users of a bank has a mean

value of 12000 and standard deviation of 2000. Calculate the proportion of customers who are spending between 8000 and 16000.

Solution. Let X be a random variable which represents the amount spent per month by a segment of credit card users of a bank. Given that $E(X) = 12000$, $\sigma = 2000$. Using Chebychev's inequality we have

$$\begin{aligned}P(8000 \leq X \leq 16000) &= P[-4000 \leq (X - 12000) \leq 4000] \\&= P[-4000 \leq (X - 12000) \leq 2 \times 2000] \\&= P[|X - 12000| \leq 2 \times 2000] \\&\geq 1 - \frac{1}{2^2} = 0.75.\end{aligned}$$

Hence the proportion of customers spending between 8000 and 16000 is at least 0.75 (or 75%).

Example 5. Suppose 1000 applications show up for a job interview, but there are only 70 positions available. To select the best 70 people amongst the 1000 applications, the employer takes an aptitude test to judge their abilities. The mean score on the test is 60, with a standard deviation of 6. If an applicant scores 84, can she/he assume that she/he is getting the job?

Solution. Let the random variable X denotes the test score. Given that, mean $(\mu) = 60$, standard deviation $(\sigma) = 6$. Using Chebychev's inequality, we have

$$\begin{aligned}P(X \geq 84) &= P[|X - 60| \geq 24] \\&\leq \frac{6^2}{24^2} = \frac{36}{576} = 0.0625.\end{aligned}$$

We can say that only $1000 \times 0.0625 = 62.5 \equiv 63$ people scored above 60. So with 70 positions available, an applicant who scores 84 can be assured that she/he will get the job.